

Education

University of Tübingen, Tübingen, Germany

October 2025 – Present

- Master of Science in Machine Learning (German GPA: 1.09/1.00, WES U.S. equivalent: 4.00/4.00)
- Studies fully funded by ELIZA's Research-Oriented Master's Scholarship.

Middle East Technical University, Ankara, Turkey

October 2020 – July 2025

- Major in Statistics & Minor in Mathematics (CGPA: 3.91/4.00) – Rank 2/140

Research Experience

EPFL, NeuroAI Lab (Lausanne, Switzerland)

Research Intern (Advisors: Yingtian Tang, Professor Martin Schrimpf)

Aug 2026 – Present

- Working on causal intervention methods for controlling representations in VLMs for neural scientific discovery.
- Linking model representations and interventions to human neural responses and comparing representations in deep networks with those in humans.

University of Tübingen, CVML Group (Tübingen, Germany)

Research Assistant (Advisors: Professor Hilde Kuehne)

Apr 2026 – Present

- Developing contrastive multimodal alignment methods for self-supervised video-language representation learning.

Middle East Technical University (Ankara, Turkey)

Research Assistant (Advisors: Dr. Jiaee Cheong, Professor Sinan Kalkan)

Aug 2024 – Sep 2026

- Designed and implemented FairSSL, a self-supervised loss for modality agnostic, fair multimodal representation learning tasks for neurological datasets (Joint work with [Cambridge Affective Intelligence and Robotics Lab](#)).
- Worked on model de-biasing strategies at the intersection of deep learning and kernel methods for fairness.

Undergraduate Researcher (Advisor: Professor Barbaros Yet)

Feb 2024 – Jan 2025

- Worked on causal agent-based models of decision theory in a medical shared decision-making setup.
- Modeled the conversation strategies employed by patients and doctors as a language game, consisting of various states, utilities, costs, and rewards.

Undergraduate Researcher (Advisor: Professor İlkey Ulusoy)

Jul 2023 – Jan 2024

- Utilized Gaussian mixture models to cluster the Inter-Subject Correlation features in fMRI and extract the segmentation map for the brain's activity region
- Worked on Dynamic Bayesian Networks developed from fMRI images to compare Alzheimer's vs Normal Brain region connectivity.

Publications and Manuscripts

- **Abtin Mogharabin**, Jiaee Cheong, Sinan Kalkan (2026), From Point to Distributional Prediction in Joint-Embedding Predictive Architectures, *Under preparation for ICLR 2027 Proceedings*
- **Abtin Mogharabin**, Jiaee Cheong, Alp Toykan Kaplan, Sinan Kalkan (2026), KINDER: Kernel-based Independence for Fair Representation Learning via Prototype-space Erasure, *Under review at NeurIPS 2026 Proceedings*
- Jiaee Cheong*, **Abtin Mogharabin***, Paul Liang, Hatice Gunes, Sinan Kalkan (2026), FairSSL: Fair multimodal self-supervised learning, *Accepted to ICML 2026 Proceedings* [[PDF](#)]
- **Abtin Mogharabin**, Vilda Purutçuoğlu. (2025), A Network Analysis of Family Dynamics, Linguistic Influences, and Happiness in Turkey, *Journal of Science: Engineering and Innovation, Gazi University, Part A, 12(3), 918-934* [[PDF](#)]
- **Abtin Mogharabin**. (2024), Modeling Emotional Functions in Medical Shared Decision-Making, preprint [[PDF](#)]

Teaching

- Student assistant for the Mathematical Statistics I and II courses (Instructor: Professor Zeynep Kalaylıoğlu) during the 2022-2023 academic year. Offered office hours for over 100 students and provided weekly individualized support and problem-solving.
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Selected Technical Projects

- Extended the experimental variable elimination library of Gen.jl probabilistic programming system (GenVariableElimination.jl) with a JAX einsum backend for scalable factor-graph contraction on GPUs, preserving modeling interface and semantics across large models with over 10x speedup on specific tests. [\[Repo\]](#)
 - Implemented and reproduced the results of Bayesian Normalization Layers (BNL), CVPR 2024, contributing to the field as no implementation was available for this paper. [\[Repo\]](#)
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Skills and Interests

- **Programming Skills:** Proficient in Python, R, and Julia. Prior experience with C++, JavaScript, MATLAB, and Haskell
 - **Machine Learning Tools:** Experienced with PyTorch, TensorFlow, PyMC, Keras, Scikit-learn, and NumPy.
 - **Miscellaneous:** Experienced in Git, UNIX, L^AT_EX, and Microsoft Excel.
 - **Languages:** Persian (Native), English (Full Professional Proficiency)
 - **Research Interests:** NeuroAI, Cognitive AI, Responsible AI, Theory of Machine Learning
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Awards and Achievements

- Accepted to EPFL's Summer@EPFL Research Internship Program (2026) at the NeuroAI lab with Professor Martin Schrimpf, selected from ~8,000 applicants (1.2% acceptance rate).
- ELIZA Excellence Research-Based Master's Scholarship: Master's studies at Tübingen fully funded by Zuse School of Excellence in Learning and Intelligent Systems (ELIZA).
- Achieved the second highest cumulative GPA in the batch of 140 graduating statistics students at METU (2025).
- METU Statistics High Honors Roll, Dean's List (2020-2025).
- Received METU's 100% international students' tuition fee payment scholarship (2021-2024).